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ANALYSIS AND DESIGN OF A ROBUST RF MEMS SWITCH

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ABSTRACT

A MEMS RF switch is expected to undergo 10 billion switching cycles before failure. Until complete physical explanation for these failure modes that include contact adhesion, damping effects, stiction, increases in resistance with time, dielectric breakdown, and electron trapping is fully established, the technology's numerous advantages cannot be harvested reliably and efficiently. This paper investigates prospective solutions to problems in switch designs by proposing a new design for the switch. We consider the new design from different perspectives: dynamic, electric, fluidic, etc. It is billed to overcome the difficulties and involves the implementation of liquid metal contact electrostatically actuated to ensure the same switching performance, with prolonged life span, and robust switching speed.

NOMENCLATURE

x Displacement variable (m)
 d Gap between upper and lower plate (m)
 m Mass (kg)
 t Time (s)
 F_i Inertial force (N)
 F_s Spring force (N)
 F_e Electrostatic force (N)
 F_d Damping force (N)
 F_p Force exerted by the droplet on the plate (N)
 ρ_z Density of zinc oxide (kg/m^3)

ρ_s Density of silicon dioxide (kg/m^3)
 L Length of Silicon dioxide upper plate (m)
 W Width of upper plate (m)
 H Thickness of upper plate (m)
 L_z Length of the conducting Tin oxide layer in upper plate (m)
 W_z Width of Tin oxide layer in upper plate (m)
 H_z Thickness of Tin oxide layer in upper plate (m)
 b Damping constant ($N.s/m$)
 μ Viscosity of air ($Pa.s$)
 r_o Radius of a perforation (m)
 r_c (Half) the separation distance between perforations (m)
 l Length of supporting beam (m)
 w Width of supporting beam (m)
 h Thickness of supporting beam (m)
 E Young's modulus (Pa)
 P Pressure (Pa)
 g_μ Gap between the droplet and the contact (m)
 V Voltage (V)
 ϵ Permittivity
 γ Surface tension (N/m)
 θ_c Contact angle (rad)

Other parameters are defined in context.

INTRODUCTION

Over the past years, radio frequency micro-electromechanical system technology has witnessed giant exciting advancements in the field of switching. The very

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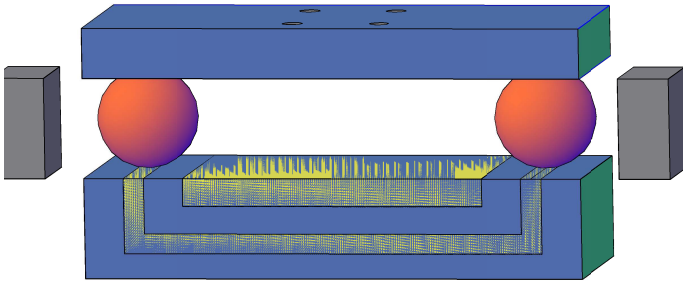


FIGURE 1. 3-D sketch of the proposed concept.

first, yet most commonly used RF MEMS switch topology, demonstrated in 1971 involves the employment of a cantilever arm to switch low-frequency electric signals [1]. Since then, several complex designs have been proposed and tested, mainly under electrostatic actuation. This explains their pervasiveness in industries spanning wireless communication, instrumentation systems, and radar systems for defence applications. In all cases, RF MEMS switches feature improvements in size [2], cost and functionality as compared to the more traditional solid state counter-parts (PIN diodes, MESFETs). Their advantage over such conventional devices is emphasized by their low power consumption, low loss performance, and many other advantages such as lack of considerable intermodulation distortion [3].

A number of challenging aspects, however, still deter their performance. To overcome these challenges scientists have developed designs that vary in the type of circuit interface (capacitive, ohmic) or in circuit configuration (shunt or series). Though in varying degrees, certain modes of failure always predominate. These include electron trapping which increases required actuation voltage, contact stiction resulting from an increase in temperature upon high frequency contact [4], dielectric breakdown, and the increase in contact resistance after a number of cycles [5]. Due to the preceding modes of failure, RF MEMS switches suffer from reliability issues that restrict their lifecycle. A simple yet effective design that extends the lifetime of the switch and eliminates many of the failure modes is proposed. The design ensures that the actuation voltage is within reasonable bounds while the switching speed is quick enough to guarantee that it continues to be useful at radio frequencies.

OPERATION MECHANISM

To overcome most problems that surface at the contact interface of the RF switch, mercury droplets will be added to this contact. Mercury is specifically used because it is the only liquid metal at room temperature that exhibits superior properties when it comes to conducting signals and surface tension. Between two

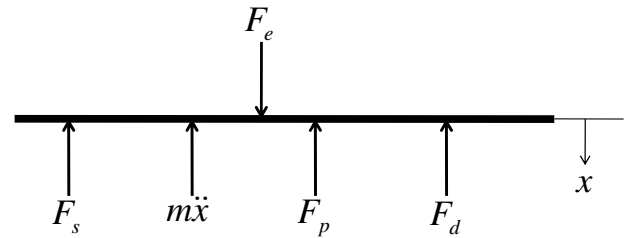


FIGURE 2. FREE BODY DIAGRAM OF ELECTROMECHANICAL EQUIVALENT SYSTEM

parallel plates, two mercury droplets will be anchored near the ends. The upper plate is insulated in the vicinity of the droplet and exposed elsewhere. The lower plate is a thick insulator (silicon dioxide) with an embedded electrode (tin oxide) that passes the signal from the one droplet to the other.

A simplified 3D sketch of the switch is given (fig. 1). The yellow hashed parts are conducting. The yellow electrodes are present on the top of the bottom plate and the bottom of the top plate, and they will cause the electrostatic force when the switch is actuated. The top plate will then bend down and squeeze the mercury, pushing it towards the side electrode and thus passing a signal through the switch as the two vertical conducting supports come in contact with the compressed mercury droplets. At this scale, the surface tension of the mercury is large enough to allow the droplet to be easily anchored and retain its shape after each compression cycle. In addition, a number of perforations are made through the plates to decrease the force due to squeezed film damping (The details of the perforation and support of the beams are not revealed in the 3D sketch).

Since the proposed system is novel in terms of its mode of operation, no detailed analytical or numerical solution of the static and dynamic performance of the above design has been reported so far. In this paper, we present an approximate analytical solution to gain physical insight of the behaviour of the entire system. We first do a rough estimation of the forces involved in order to size the switch. Static analysis depicts the required actuation voltage at varying starting upper plate locations. Dynamic analysis, which takes into account energy considerations of the switch, determines the switching specs of the device. All throughout the analyses, upper plate movement is approximated as one dimensional.

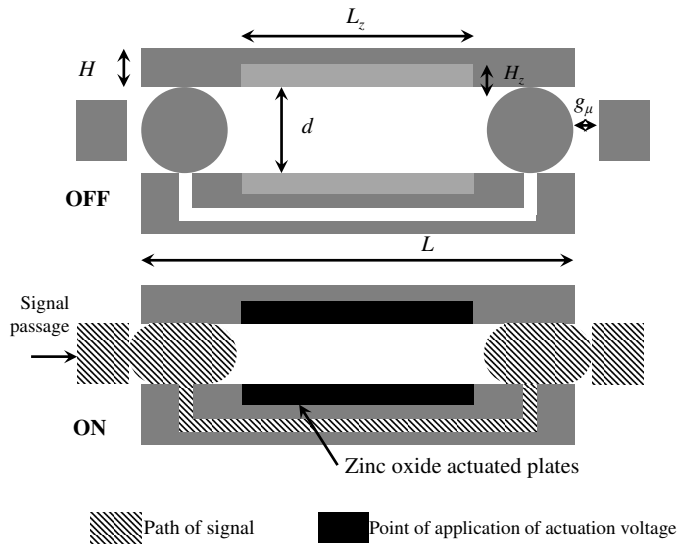


FIGURE 3. TOP: 2-D SKETCH OF THE SWITCH SHOWS 2 MERCURY DROPLETS ELECTRICALLY CONNECTED BY AN EMBEDDED ELECTRODE (SHOWN AS THE WHITE CHANNEL IN TOP FIGURE). BOTTOM: SWITCH DURING THE 'ON' STATE.

MODELLING AND DERIVATION OF AN EQUATION OF MOTION

Quantifying the Forces

Several forces act on the upper plate as it is lowered under the effect of the electrostatic actuation. For the sake of our paper, the imposed forces will be modelled as a spring or a dashpot depending on the phenomena dictating their behaviour, the spring being a model of the elastic property of the beams and the dashpot that of the squeezed film damping effect. This will help determine the magnitude of each of the existing forces at any intermediate position of the upper-plate following actuation, which will facilitate simulations and subsequent testing. Several forefront assumptions include that the upper beam is a spring-less mass connected to a mass-less spring (supporting beams). The following forces act on the upper plate (fig. 2):

1. Force by the supporting beams on the upper plate (F_s). This is modelled as a spring force. The stiffness of the supporting beams (linked to their material) is the principal factor determining its importance.
2. Electrostatic force (F_e) due to applied actuation voltage. The actuator, coupled with the upper conducting surface of the upper plate, forms a parallel plate capacitor. The air gap separating the two plates acts as a dielectric layer. When a difference in potential is created between the actuator and the upper plate, an electrostatic force is induced, leading to the movement of the upper plate.
3. Pressure force (F_p) due to surface tension of the droplet.



FIGURE 4. FIXED-FIXED FLEXURE OF THE UPPER PLATE (TOP VIEW OF THE PLATE).

Squeezing the mercury droplet (modelled as an incompressible fluid) causes a pressure difference to develop across surface of droplet. This increase in pressure is due to the surface tension of mercury.

4. Force by air on the upper plate (F_d). This is modelled as a damping force since the air is modelled as a dashpot that constrains the movement of the upper plate (damping). The squeeze film damping is the predominating phenomenon exhibited by the air as the upper plate is lowered to its final position.
5. Inertial force

Inertial Force While under displacement, the upper plate develops an inertial force demonstrated by its resistance to the change of its state. This force is given by $m\ddot{x}$ where m is the total mass of the upper plate and $\ddot{x}$ is its acceleration. As shown in figure 2, the upper plate is composed of two layers, one of which is silicon dioxide that acts as an insulator and has great flexibility and the other is Tin Oxide that acts as a conductor. Choice of material takes into account various design specifications that aim at yielding optimum switching behaviour.

Hence, the force of inertia is given as a function of material densities and plate dimensions as

$$[\rho_z L_z W_z H_z + \rho_s (LWH - L_z W_z H_z)] \ddot{x} \quad (1)$$

Damping Term As the upper plate bends towards the lower one, the drop compresses and the air in between starts to flow out in from both directions. Since this flow is viscous, some of the kinetic energy of the moving plate gets dissipated, hence the squeeze film damping effect. This also applies when the two surfaces move apart; a case in which the fluid has to rush into the expanding space between the surfaces [3]. A number of assumptions are to be made before deriving the damping force:

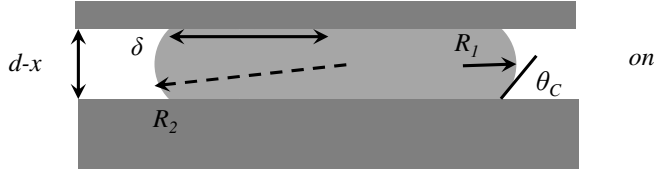


FIGURE 5. SURFACE TENSION ILLUSTRATION FOR ONE DROPLET IN THE ON AND OFF SWITCH STATES.

- The gap is much smaller than the dimensions of the plate
- The plate can be modelled as infinitely long based on our initial sizing of the switch

The damping force is thus written as [6]

$$F_d = L_s W_s \frac{3\mu r_c^2}{2x^3} \dot{x} \left(4 \left(\frac{r_o}{r_c} \right)^2 - \left(\frac{r_o}{r_c} \right)^4 - 4 \ln \left(\frac{r_o}{r_c} \right) - 3 \right) \quad (2)$$

Spring Force For the fixed-fixed flexure configuration composed of four beams, the linear elastic stiffness due to bending is given by [7]

$$F_s = \left[4Ew \left(\frac{h}{l} \right)^3 \right] x \quad (3)$$

Electrostatic Force The attractive electrostatic force is the driving force behind the switch. It written as follows [3]

$$F_e = \frac{\epsilon L_z W_z V^2}{2(d-x)^2} \quad (4)$$

Equation of Motion The equation of motion is written as

$$[\rho_z L_z W_z H_z + \rho_s (LWH - L_z W_z H_z)] \ddot{x} + \left[4Ew \left(\frac{h}{l} \right)^3 \right] x + L_s W_s \frac{3\mu r_c^2}{2x^3} \dot{x} \left(4 \left(\frac{r_o}{r_c} \right)^2 - \left(\frac{r_o}{r_c} \right)^4 - 4 \ln \left(\frac{r_o}{r_c} \right) - 3 \right) + \gamma \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \pi \delta^2 = \frac{\epsilon L_z W_z V^2}{2(d-x)^2} \quad (11)$$

Analysis of the Derived Reduced Order Model

Static Analysis

The power required to actuate the proposed switch is one of the main parameters that must be determined. This voltage de-

Surface Tension Force Based on the Young-Laplace relation, balancing the tension forces with pressure leads

$$\Delta P = \gamma \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \quad (5)$$

Where R_1 and R_2 are the radii of curvature in each of the axes that are parallel to the surface. By projecting the radii, we get (θ being the contact angle):

$$\delta = \sqrt{\frac{d^3 - (d-x)^3}{3(d-x)}} \quad (6)$$

$$R_1 = \frac{-(d-x)}{2 \cos \theta_C}, \cos \theta_C < 0 \text{ for mercury} \quad (7)$$

$$R_2 = \delta + R_1 (1 - \sin \theta_C) \quad (8)$$

and finally, the force is quantified as

$$F_p = \Delta P A = \gamma \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \pi \delta^2 \quad (9)$$

We may also write

$$g_\mu = \sqrt{\frac{(d^3 - (d-x)^3)}{3(d-x)}} + \frac{x(1 - \sin \theta_C)}{(2 \cos \theta_C)}; \quad (10)$$

The preceding analysis is done under the assumptions that the curvature is fully circular, the flow is axisymmetric, the pressure outside the droplet is near atmospheric, and the pressure on the other side of the plate is atmospheric.

termines whether such a device is practical, by allowing a mode of direct comparison across other similar MEMS devices whose actuation voltages are already well documented. To approximate the switching voltage, static analysis was performed. From this

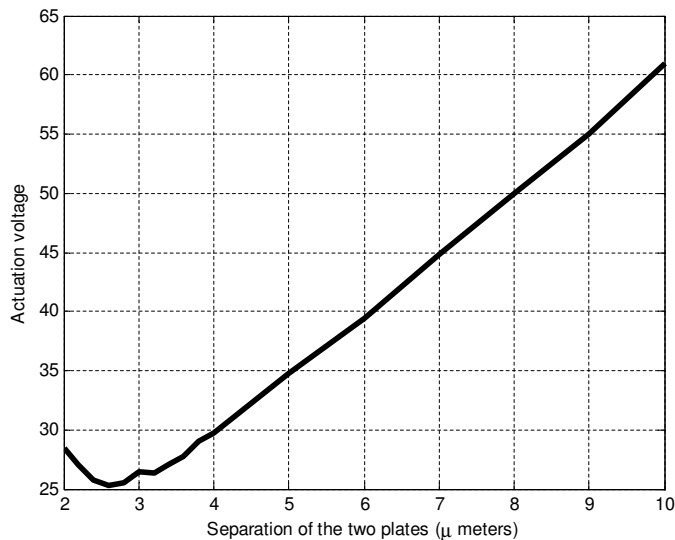


FIGURE 6. THE ACTUATION VOLTAGE REQUIRED TO SQUEEZE A DROPLET OF SPECIFIC DIAMETER.

analysis, the optimum droplet size was determined such that the required actuation voltage was minimal. One important constraint was imposed on our analysis: g_μ (the mercury-vertical support clearance) is at least $0.5 \mu m$ (for manufacturing purposes). The initial analysis took into account the frequency-independent forces only. During this analysis, the plates' size was chosen, the gap was determined (fig. 6), and the perforation effect were most easily isolated (fig. 7). The minimal input voltage occurs at a droplet diameter of $3.19 \mu m$ (figure 6). This will be the droplet diameter used in later simulations. For diameters smaller or larger than that, the performance of the switch drops. The damping effects are also seen in figs. 8 & 9.

Dynamic Analysis

In order to fully understand the motion of the plates, several other parameters need to be investigated. These parameters include the rise time (the time for the moveable upper plate to go from its initial position to the final position when the mercury droplet is first in contact with the vertical plate). Once determined, these two parameters help characterize the maximum frequency of the signal. It is also possible to ascertain a better estimate of the actual voltage using this model as compared to static analysis. In order to estimate the fall and rise time, a plot of displacement versus time is required. The system's equation of motion is highly non-linear and was thus simulated using SIMULINK software.

An iterative procedure was sought to attain the best combination of the above parameters. Changing one variable, while holding all other variables constant, enabled us to determine the

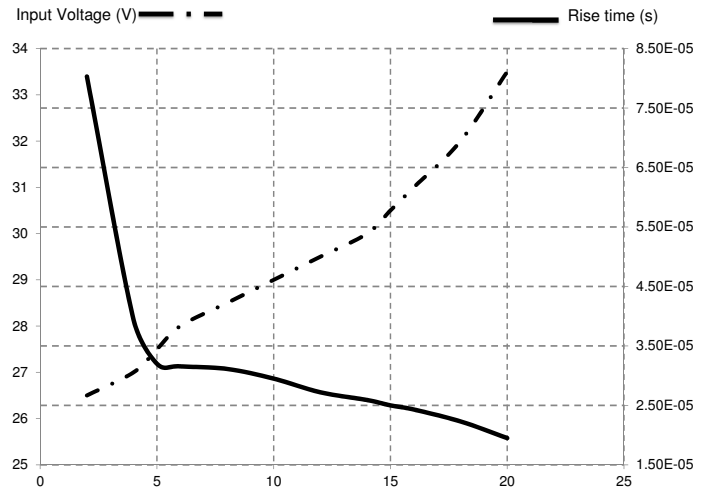


FIGURE 7. DEPENDENCE OF INPUT VOLTAGE AND RISE TIME ON % PERFORATION.

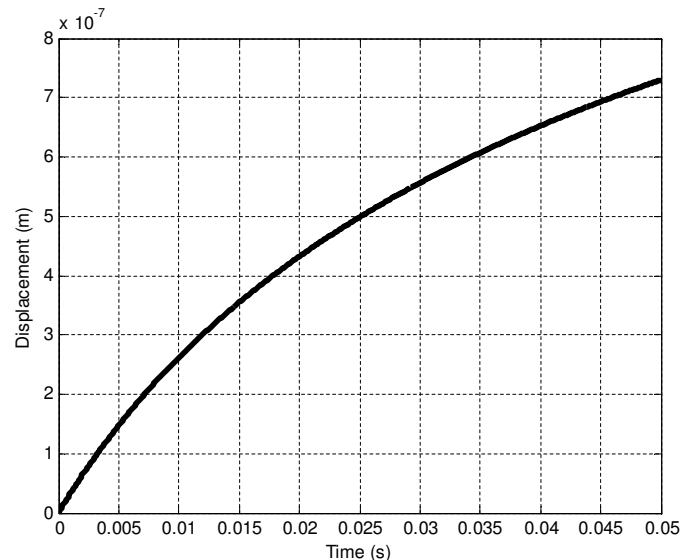


FIGURE 8. DISPLACEMENT VS TIME FOR NON-PERFORATED CONFIGURATION AT 99 % DUTY CYCLE AND 26.3 V.

effect of each variable on performance of the system. Factors investigated included the stiffness of the supporting beams, the damping coefficient due to squeeze film damping, and the geometry of the switch (area of plate, gaps between the mercury droplet). Based on above procedure, it was found that the values of the fall and rise time were most heavily dependent on the damping coefficient. In order to achieve reasonable results for fall and rise time, a low damping coefficient was required. To reduce this damping coefficient, without compromising on size

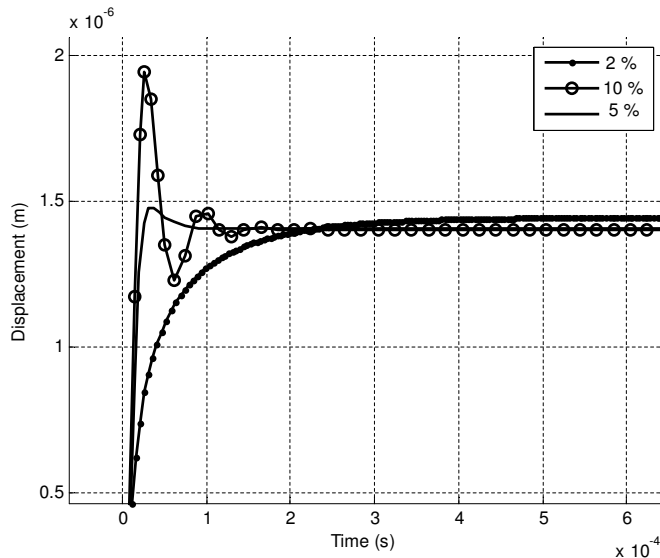


FIGURE 9. DISPLACEMENT VS TIME FOR PERFORATED CONFIGURATIONS AT 99 % DUTY CYCLE AND 26.3 V.

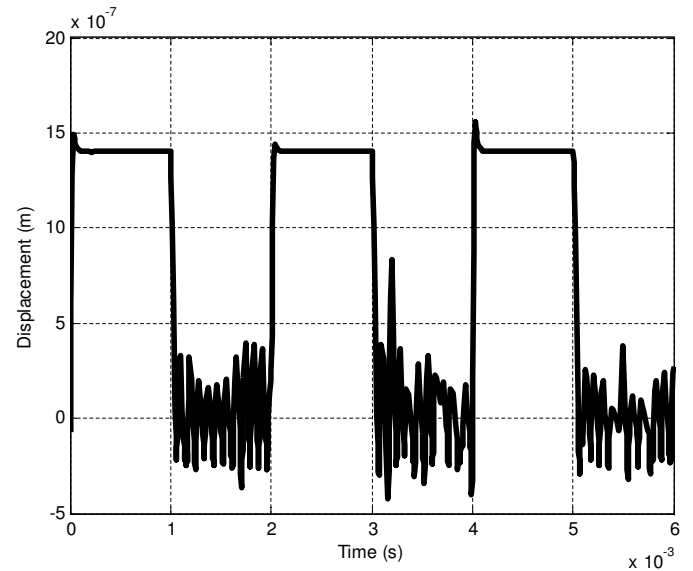


FIGURE 11. DISPLACEMENT-TIME CURVE (FOR 3 PERIODS) AT 26.3 V WITH 5 % PERFORATION AND 50 % DUTY CYCLE.

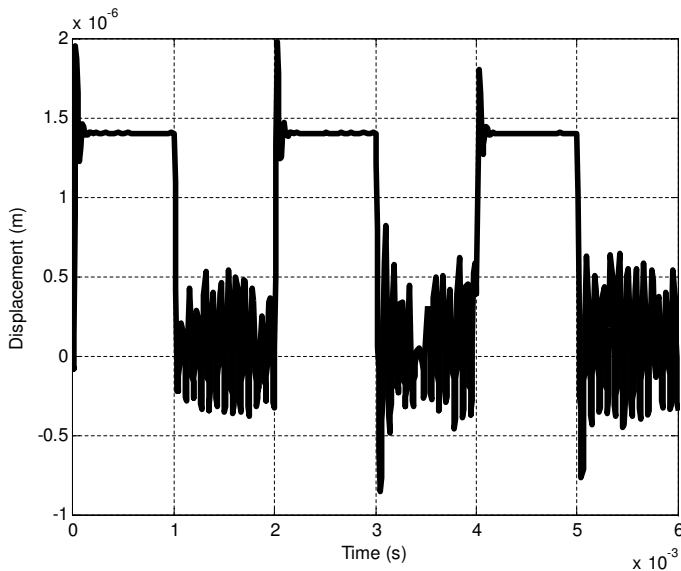


FIGURE 10. DISPLACEMENT-TIME CURVE (FOR 3 PERIODS) AT 26.3 V WITH 10 % PERFORATION AND 50 % DUTY CYCLE.

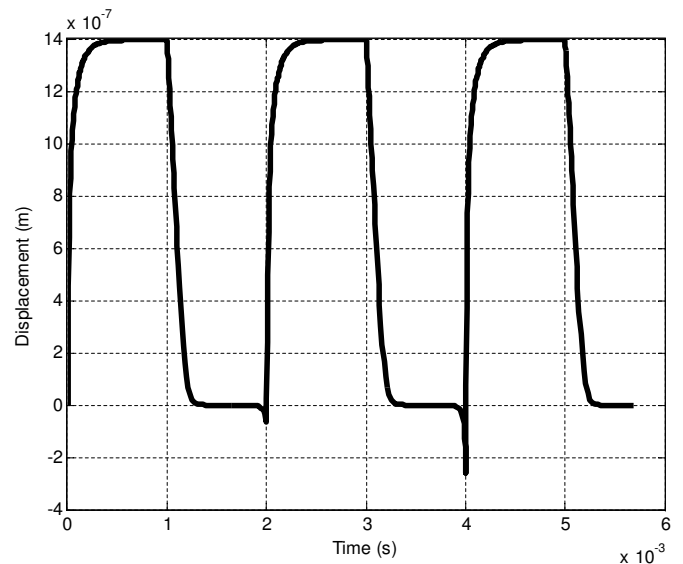


FIGURE 12. DISPLACEMENT-TIME CURVE (FOR 3 PERIODS) AT 26.5 V WITH 2 % PERFORATION AND 50 % DUTY CYCLE.

requirements of RF MEMS switches, the option of perforating the plate was studied. This is revealed by comparing figures 10, 11, and 12. The well-behaved behaviour occurs at 2 percent perforation. For non-perforated plates, it was determined that the voltage required (and time-dependent behaviour) were both inadequate (fig. 7). For 5 and 10 % perforation, we have an under-damped system.

CONCLUDING REMARKS

The novel design proposed above offers MEMS switching technologies the potential to address and minimize the prevalent modes of failure of increase in contact resistance, stiction, squeeze film damping, dielectric breakdown, and pull-in. By replacing the solid contacts with liquid mercury, we are able to circumvent the issues resulting from the dynamic strain at high

Parameter	Proposed Design
Voltage (V)	26.3
Current (A)	1.0168×10^{-8}
Switching Speed (μ s)	880
Power (W)	2.674×10^{-7}

TABLE 1. OPERATING PARAMETERS OF THE PROPOSED DESIGN.

d	Gap between upper and lower plates	$3.19 \mu m$
	Maximum displacement	$1.4 \mu m$
L	length of the upper plate	$500 \mu m$
W	width of the upper plate	$500 \mu m$
H	thickness of the upper plate	$2 \mu m$
L_z	length of the conducting part of upper plate	$400 \mu m$
W_z	width of the conducting part of upper plate	$400 \mu m$
H_z	thickness of the conducting part of upper plate	$1 \mu m$
l	length of the supporting beam	$500 \mu m$
w	width of the supporting beam	$10 \mu m$
h	height of the supporting beam	$5 \mu m$
g_μ	gap between the droplet and the contact	$3.7 \mu m$

TABLE 2. DIMENSIONS OF THE PROPOSED DESIGN.

frequency switching. This configuration also helps with the dielectric breakdown and increased pull in voltage as the droplet is carefully sized to produce a satisfactory mechanical performance with the required electrical characteristics in check.

This design makes the switch more stable since the droplet acts a barrier to the locking of the switch. These factors com-

bined allow us to believe that the proposed design will have a longer lifetime than current existing MEMS switches. The proposed design also falls within other RF MEMS switches in terms of performance parameters (table 1) [7]. The results in table 2 are attained at the optimized values of different parameters.

A quantification of the predicted lifetime will be possible only after further physical experimentation. Future work should involve such experimental investigation of the design, in addition to looking into the substitution of mercury with an alternative to better comply with the European stance on its use.

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